



**Riphah International University**  
Final Year Project Proposal

## **Aurum AI**

### **AI-Powered Gold Price Forecasting & Market Intelligence Platform**

#### **Project Team**

Mir Ahmad Shah (54906)

Shahid Zaman (55123)

#### **Supervised By**

Mr. Muhammad Shabbir Hassan (Senior Lecturer)

## Table of Contents

|                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| <b>1. Introduction and Background .....</b>                                | <b>3</b>  |
| <b>How the Problem Developed .....</b>                                     | <b>3</b>  |
| <b>2. Literature Review .....</b>                                          | <b>3</b>  |
| <b>3. Gap Analysis .....</b>                                               | <b>4</b>  |
| <b>4. Problem Statement .....</b>                                          | <b>5</b>  |
| <b>5. Proposed Solution and Methodology.....</b>                           | <b>5</b>  |
| <b>5.1 System Architecture and Main Outputs .....</b>                      | <b>6</b>  |
| <b>5.2 Machine Learning Methodology .....</b>                              | <b>6</b>  |
| <b>5.3 Event-Aware Analysis and User Support .....</b>                     | <b>7</b>  |
| <b>5.4 Sentiment Analysis for Global-Event Impact on Gold Prices .....</b> | <b>7</b>  |
| <b>6. Project Scope .....</b>                                              | <b>8</b>  |
| <b>7. Roles and Responsibilities.....</b>                                  | <b>9</b>  |
| <b>8. Constraints and Limitations .....</b>                                | <b>9</b>  |
| <b>8.1 Ethical and Disclaimer Considerations.....</b>                      | <b>10</b> |
| <b>9. Expected Outcomes.....</b>                                           | <b>10</b> |
| <b>10. Complex Computing Problem Analysis .....</b>                        | <b>11</b> |
| <b>11. Conclusion .....</b>                                                | <b>13</b> |

## 1. Introduction and Background

---

Gold has long been regarded as a safe-haven asset that investors turn to during periods of economic uncertainty. Its price, however, can move quickly whenever market conditions shift, and retail users frequently lack access to timely, evidence-based information to interpret those movements. Advanced market analysis tools that could help are often difficult to access or prohibitively expensive for individual users, leaving a gap between the information available to institutional investors and that available to the general public.

Aurum AI is proposed to close this gap. The platform will bring together live gold prices and AI-based market analysis into a single dashboard, helping users understand market conditions without the system itself executing trades on their behalf. The platform's initial focus is on COMEX Gold Futures (GC=F) and Spot Gold (XAU/USD), the two most widely tracked gold price benchmarks.

### How the Problem Developed

Academic research has progressively moved from traditional statistical analysis toward AI-based approaches for gold-price forecasting, including neural networks and hybrid models. More recent studies have expanded this further by incorporating technical indicators, ensemble methods, and additional market variables. Despite this progress, much of the existing research addresses only one part of the problem at a time forecasting or sentiment, for example rather than treating it as an integrated system.

Aurum AI proposes a single platform that connects data collection, AI-based analysis, alerting, and user support in one coherent workflow. In addition to price history, the platform incorporates broader market context such as sentiment, the VIX volatility index, interest rates, and oil prices, recognizing that gold forecasting can benefit from signals beyond price history alone.

## 2. Literature Review

---

The table below summarizes key studies that inform the design of Aurum AI, spanning neural-network forecasting, hybrid volatility models, ensemble classifiers, and sentiment-based approaches to market analysis.

| Sr. # | Study                            | Method                              | Main Finding                                                      | Limitation                                    |
|-------|----------------------------------|-------------------------------------|-------------------------------------------------------------------|-----------------------------------------------|
| 1     | Li & Ma (2025)                   | News uncertainty                    | News-related uncertainty can help explain gold volatility         | Focuses on volatility, not full user workflow |
| 2     | Cohen & Aiche (2023)             | ML with VIX, yields and market data | Broader financial variables can improve gold forecasting          | More data sources increase complexity         |
| 3     | Kristjanpoller & Minutolo (2015) | ANN-GARCH                           | Hybrid modeling can improve gold volatility forecasts             | More complex than a simple ML pipeline        |
| 4     | Ballings et al. (2015)           | Ensemble classifiers                | Ensembles can improve prediction compared with some single models | Stock direction study, not gold specific      |
| 5     | Hutto & Gilbert (2014)           | VADER sentiment                     | Fast, rule-based sentiment scoring for short text                 | General sentiment model, not gold specific    |
| 6     | Bollen et al. (2011)             | Twitter mood                        | Public mood can contain market information                        | Social media data can be noisy                |
| 7     | Parisi et al. (2008)             | Rolling / recursive neural networks | Dynamic neural models can forecast gold price direction           | Older study; not a deployed platform          |

*Table 1: Summary of reviewed literature relevant to gold-price forecasting and market-sentiment analysis.*

### 3. Gap Analysis

The literature reveals several recurring gaps between academic research and practically usable, deployed forecasting systems. Aurum AI is designed specifically to close each of these gaps, as summarized below.

| Sr. # | Gap        | Current Situation                       | Aurum AI Proposal           |
|-------|------------|-----------------------------------------|-----------------------------|
| 1     | Deployment | Many studies remain offline experiments | Develop a live web platform |

| Sr. # | Gap                    | Current Situation                                                                              | Aurum AI Proposal                            |
|-------|------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------|
| 2     | Time-series validation | Some time series studies use random train/test splitting, which can introduce temporal leakage | Use walk-forward expanding-window validation |
| 3     | Baseline comparison    | Simple baselines are not always reported                                                       | Compare models with a persistence baseline   |
| 4     | Model updating         | Models may be trained only once                                                                | Schedule retraining and re-evaluation        |
| 5     | Quality control        | Limited rules for accepting new models                                                         | Use champion / challenger checks             |
| 6     | Market context         | Some approaches use mainly gold history                                                        | Include VIX, DXY, Treasury yield and oil     |
| 7     | Event awareness        | Historical prices may miss sudden shocks                                                       | Use Safety Gate and news sentiment           |
| 8     | User accessibility     | Research outputs may be difficult for non-technical users                                      | Provide a simple dashboard and support layer |

*Table 2: Identified gaps in existing research and tools, and how Aurum AI addresses them.*

## 4. Problem Statement

This project addresses three practical problems in the gold-market analysis space:

- Problem 1: Forecasting: Retail users do not have easy access to scientifically evaluated short- and medium-term gold forecasts.
- Problem 2: Volatility: Historical price patterns alone may not explain sudden changes caused by news or broader market conditions.
- Problem 3: Events: Large gold-price movements can happen quickly; without automatic alerts, users may notice important changes too late.

## 5. Proposed Solution and Methodology

Aurum AI proposes an integrated system spanning data ingestion, feature engineering, machine-learning forecasting, event-aware safety checks, and a user-facing dashboard. The following subsections describe the system architecture, the machine-learning methodology, the event-

aware analysis layer, and the sentiment-analysis component used to capture the effect of global events on gold prices.

### **5.1 System Architecture and Main Outputs**

The system architecture transforms raw market data into user-facing results through a structured pipeline: data collection, feature engineering, model inference, safety checking, and presentation. The platform will deliver the following main outputs to users:

- Live prices and charts for GC=F (COMEX Gold Futures) and XAU/USD (Spot Gold)
- A 30-day machine-learning forecast
- A BUY / SELL / HOLD technical signal
- Volatility email alerts
- An UNCERTAIN status shown during unusual market conditions
- A user profile area and a support chatbot

### **5.2 Machine Learning Methodology**

The forecasting engine is built on a causal feature set that is, features computed only from information available at the time of prediction, to avoid look-ahead bias. This feature set includes:

- 7 return lags (past price returns over different time periods)
- 4 rolling statistics (recent average returns and volatility)
- 4 mean-reversion features (current price compared with moving averages)
- 1 RSI feature (recent buying and selling strength)
- 3 momentum features (strength and direction of price movement)
- 4 seasonality features (monthly and weekly time patterns)
- 4 volatility features (size and risk of price movements)
- 4 OHLC features (daily Open, High, Low, Close structure)

Model validation follows 4 expanding walk-forward folds with no random shuffling, and a minimum of 252 training bars (approximately one year) to reflect a realistic, time-respecting deployment scenario. Candidate models include XGBoost, LightGBM, and LSTM/GRU architectures, each compared against a persistence baseline. Evaluation metrics include MAE, directional accuracy, precision, recall, F1-score, and the confusion matrix. A predefined

performance threshold determines whether a model is suitable for deployment, and the best-performing model or ensemble is selected through walk-forward validation.

### 5.3 Event Aware Analysis and User Support

Beyond price-based modelling, Aurum AI incorporates four complementary components designed to make the platform aware of real-world context and accessible to non-technical users:

- **Safety Gate:** detects unusual market conditions, checks recent volatility, and returns an UNCERTAIN / LOW CONFIDENCE status when reliability is reduced, with the goal of reducing false confidence during market shocks.
- **Macro Context:** incorporates the VIX (market anxiety and uncertainty), DXY (US Dollar Index), the US 10-Year Treasury yield, and crude oil prices, to add broader market context beyond gold price history alone.
- **News Sentiment:** analyses gold-related headlines to include event information alongside price data (detailed further in Section 5.4).
- **Support Chatbot:** provides knowledge-based help covering forecast meaning, signal meaning, alert settings, and dashboard guidance, aimed at helping non-technical users interpret the platform correctly.

### 5.4 Sentiment Analysis for Global-Event Impact on Gold Prices

Gold prices are strongly influenced by global events geopolitical tensions, central-bank interest-rate decisions, inflation data releases, conflicts, elections, and macroeconomic shocks that price history alone cannot anticipate. A model trained only on historical prices will always be reactive to these shocks rather than responsive to the information that precedes them. To address this directly, Aurum AI will use sentiment analysis as a core input for predicting and explaining gold-price movements that are driven by global events.

The sentiment-analysis component will apply VADER (Valence Aware Dictionary and sentiment Reasoner), a lexicon- and rule-based sentiment scoring method well suited to short-form financial headlines, to a continuous stream of gold-related news. Specifically, the system will compute:

- **1-hour sentiment:** a short rolling window capturing the market's immediate reaction to breaking news and unfolding global events.
- **24-hour sentiment:** a longer rolling window capturing sustained sentiment shifts as an event continues to develop or is digested by the market.

- News volume: the number of gold-related headlines in a given window, used as a proxy for how significant or fast-moving an event is.

These sentiment features are fused with the technical, macro, and volatility features described in Section 5.2, so that the forecasting models can adjust their outputs when global events are actively shaping market sentiment rather than relying on price history alone. When sentiment shifts sharply or news volume spikes beyond expected levels, this information also feeds into the Safety Gate (Section 5.3): the platform will flag its forecasts as UNCERTAIN / LOW CONFIDENCE rather than presenting a false sense of precision during periods of high event-driven uncertainty.

In this way, sentiment analysis becomes the mechanism through which Aurum AI predicts and communicates how global events affect the price of gold: it quantifies real-world events as they are reported in the news, links that quantified signal to the forecasting pipeline, and ensures that users are warned rather than misled when unfolding events make short-term prediction unreliable. This directly targets Problem 3 (Events) identified in Section 4, and closes the “Event awareness” gap identified in Section 3.

## 6. Project Scope

The functional scope of Aurum AI covers the following capabilities:

- GC=F and XAU/USD monitoring
- 30-day ML forecasting
- BUY / SELL / HOLD technical signals
- Volatility email alerts
- Walk-forward validation and model quality checks
- VIX, DXY, Treasury yield and oil context
- News sentiment and volume analysis
- Safety Gate and support chatbot

| Sr. # | In Scope          | Out of Scope       |
|-------|-------------------|--------------------|
| 1     | Gold forecasting  | Trade execution    |
| 2     | Market monitoring | Broker integration |

| Sr. # | In Scope             | Out of Scope         |
|-------|----------------------|----------------------|
| 3     | Alerts               | Portfolio management |
| 4     | Event-aware analysis | Other commodities    |
| 5     | User dashboard       | Financial advice     |

Table 3: In-scope and out-of-scope items for the Aurum AI project.

## 7. Roles and Responsibilities

Responsibilities have been planned across the two team members as follows; these are planned allocations and can be adjusted during development.

| Sr. # | Role                 | Main Responsibility                                  | Assigned To    |
|-------|----------------------|------------------------------------------------------|----------------|
| 1     | Project Coordination | Planning, integration and documentation              | Both           |
| 2     | ML / Data            | Data pipeline, features, training and evaluation     | Mir Ahmad Shah |
| 3     | Backend / DevOps     | Flask API, cloud services, automation and deployment | Mir Ahmad Shah |
| 4     | Frontend / UX        | React dashboard, authentication, charts and UX       | Shahid Zaman   |
| 5     | Testing / QA         | Functional testing, model evaluation and evidence    | Shahid Zaman   |

Table 4: Planned division of responsibilities between team members.

## 8. Constraints and Limitations

| Sr. # | Area           | Constraint / Limitation                 | Planned Mitigation                     |
|-------|----------------|-----------------------------------------|----------------------------------------|
| 1     | Infrastructure | Free-tier RAM, storage and usage limits | Caching and efficient queries          |
| 2     | APIs           | Rate limits and source availability     | Multiple sources and fallback strategy |

| Sr. # | Area                 | Constraint / Limitation                             | Planned Mitigation                            |
|-------|----------------------|-----------------------------------------------------|-----------------------------------------------|
| 3     | Hardware             | No GPU available                                    | CPU-friendly tree-based models                |
| 4     | Data Resolution      | Daily data does not capture every intraday movement | State the limitation clearly to users         |
| 5     | Unknown Events       | Future shocks cannot be predicted reliably          | Safety Gate and event-aware inputs            |
| 6     | Forecast Uncertainty | Point forecasts do not show full uncertainty ranges | Consider uncertainty estimation in evaluation |
| 7     | Project Scope        | Two-member FYP with limited time                    | Prioritize the planned workflow               |

*Table 5: Anticipated technical, data, ML, and project constraints and their planned mitigations.*

### 8.1 Ethical and Disclaimer Considerations

The platform's handling of user data and financial-risk communication will follow these principles:

- Predictions will be shown as AI-generated estimates, not guaranteed results.
- The system will not execute trades or act as a broker.
- The Safety Gate will communicate when model reliability is reduced.
- User email data will be used for planned alerts and stored securely.
- The system is intended for research and education, not licensed financial advice.
- Users remain responsible for their own financial decisions.

**Disclaimer:** Aurum AI is intended to support market analysis and learning; it does not guarantee profit or investment success.

## 9. Expected Outcomes

By the end of the FYP, the project is expected to deliver the following:

### Platform

A deployable web application with live gold data, forecasting, technical signals, alerts, profile settings, and user support.

### ML Pipeline

A documented forecasting workflow with causal features, walk-forward validation, baseline comparison, and model-quality metrics.

### Automation

Scheduled training and evaluation with model-quality checks and controlled promotion of better models.

### User Safety

Clear communication of uncertainty, market shocks, and the limits of AI-based forecasting.

## 10. Complex Computing Problem Analysis

Final Year Projects in computing disciplines are expected to demonstrate engagement with a **complex computing problem**, as defined by the Seoul Accord criteria. The reference table below sets out the nine characteristics used to judge whether a problem qualifies as complex.

The following table maps each of the nine Seoul Accord characteristics to the specific way in which Aurum AI satisfies it, demonstrating that the project qualifies as a complex computing problem rather than a routine software build.

| Sr.# | Characteristic                    | Seoul Accord Description                                                                                                 | How Aurum AI Satisfies It                                                                                                                                                                                                                                                                                                                  |
|------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Range of conflicting requirements | Involves wide ranging or conflicting technical, computing, and other issues.                                             | Aurum AI must balance conflicting requirements: real-time responsiveness vs. free-tier infrastructure limits, forecasting accuracy vs. model latency and interpretability, and combining heterogeneous data (price, macro indicators, and news sentiment) into one coherent pipeline without any single data source dominating the result. |
| 2    | Depth of analysis required        | Has no obvious solution, and requires conceptual thinking and innovative analysis to formulate suitable abstract models? | There is no ready-made template for fusing causal price features, macro-economic context, and news sentiment into a single walk-forward-validated forecasting and alerting system. The project requires original                                                                                                                           |

| Sr.# | Characteristic                                                          | Seoul Accord Description                                                                                                           | How Aurum AI Satisfies It                                                                                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                         |                                                                                                                                    | analysis to design the feature set, the Safety Gate logic, and the champion/challenger model-promotion strategy.                                                                                                                                                                                           |
| 3    | Depth of knowledge required                                             | A solution requires the use of in-depth computing or domain knowledge and an analytical approach based on well-founded principles. | The project draws on machine learning (XGBoost, LightGBM, LSTM/GRU), time-series validation theory, financial-market domain knowledge (gold, VIX, DXY, yields, oil), natural-language sentiment analysis (VADER), and full-stack software engineering (Flask, React, cloud deployment).                    |
| 4    | Familiarity of issues                                                   | Involves infrequently encountered issues.                                                                                          | Combining event-aware safety gating with a live, retrainable ML forecasting service and reliably distinguishing genuine model uncertainty from ordinary market noise is a design problem the team has not previously encountered in coursework or standard projects.                                       |
| 5    | Level of problem                                                        | Are outside problems encompassed by standards and standard practice for professional computing.                                    | There is no standard, off-the-shelf procedure for deploying a live, walk-forward-validated, sentiment- and macro-aware gold-forecasting platform with automated retraining and quality gating; the team must design and justify its own validation, deployment, and safety-gating standards.               |
| 6    | Extent of stakeholder involvement and level of conflicting requirements | Involves diverse groups of stakeholders with widely varying needs.                                                                 | Stakeholders include non-technical retail users needing simple guidance, the academic supervisor evaluating technical rigor, and the implicit expectations of financial-ethics and regulatory norms requiring the system to be simultaneously accessible, technically sound, and responsibly communicated. |
| 7    | Consequences                                                            | Has significant consequences in a range of contexts.                                                                               | Because forecasts and BUY/SELL/HOLD signals can influence users' real financial decisions, incorrect or overconfident predictions carry real consequences which is why the Safety Gate, disclaimers, and uncertainty communication are treated as first-class                                              |

| Sr.# | Characteristic             | Seoul Accord Description                                                             | How Aurum AI Satisfies It                                                                                                                                                                                                                                                                                                                         |
|------|----------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                            |                                                                                      | design requirements rather than afterthoughts.                                                                                                                                                                                                                                                                                                    |
| 8    | Interdependence            | Is a high-level problem possibly including many component parts or sub problems?     | The platform decomposes into many interdependent sub-problems: data ingestion, feature engineering, model training and validation, sentiment analysis, macro-context integration, the Safety Gate, alerting, a React dashboard, authentication, a support chatbot, and automated retraining each of which must integrate cleanly with the others. |
| 9    | Requirement identification | Identification of a requirement or the cause of a problem is ill-defined or unknown. | What counts as an “unusual market condition,” an acceptable performance threshold for deployment, or a meaningful sentiment/news-volume spike is not given in advance; these thresholds are ill-defined at the outset and must be identified and justified through the team's own research, experimentation, and the gap analysis in Section 3.   |

*Table 6: Mapping of Aurum AI's design decisions to the Seoul Accord complex-computing-problem criteria.*

## 11. Conclusion

Aurum AI proposes an integrated, research-grounded platform that brings together causal machine-learning forecasting, macro-economic context, event-aware sentiment analysis, and a safety-first user experience for retail gold-market participants. By combining walk-forward-validated forecasting models with a dedicated sentiment analysis pipeline for global events, the platform is designed to predict and explain gold-price movements more responsively than price-history-only approaches, while remaining transparent about the limits of AI-based prediction.

The project satisfies the recognized characteristics of a complex computing problem through its conflicting technical requirements, its need for original analysis and deep domain knowledge, its many interdependent sub-systems, and the real-world consequences of its outputs. Taken together, the proposed solution, methodology, and safeguards described in this report form a coherent and achievable plan for the Final Year Project.